

Financial Distress and Tail Risk

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Abstract

Considering the declining number of bankruptcy filings, and increasing out-of-court negotiations and debt reorganizations, we argue in favour of penalizing firms for becoming sufficiently close to bankruptcy that they have questionable going-concern status. Thus, we propose a definition of financial distress contingent upon firms' earnings, financial expenses, market value and operating cash flow. Subsequently, we investigate the role of tail risk measures (Value-at-risk and Expected Shortfall) in aggravating likelihood of financial distress. Our results show that longer horizon (three- and five-year) tail risk measures contributes positively toward firms' likelihood of entering financial distress.

Keywords: tail risk; value at risk; downside risk; expected shortfall; credit risk; financial distress

JEL Classification Codes: G17, G33, C53

1. Introduction

Financial Distress and *Tail Risk* are two apparently diverse topics that are gaining increasing attention in corporate finance literature. The financial crisis of 2007-08 was the alarm bell that augmented global awareness toward tail risk among financial risk managers. Since then, we have witnessed increasing concern among stakeholders toward firms' risk of bankruptcy or financial distress. Although tail risk has been an active area of investigation in the domain concerned with large financial institutions and financial stability, to the best of our knowledge this study is the first academic attempt to address the relationship between firms' extreme negative daily equity returns and their likelihood of experiencing financial distress. We hypothesize that more frequent extreme negative daily equity returns result in higher tail risk, and this subsequently increases firms' likelihood of entering financial distress.

The vast majority of academic literature on bankruptcy prediction gravitates around either the choice of explanatory variables (e.g. Campbell *et al.*, 2008; Jones 2017), or modelling methodologies (e.g. Shumway, 2001; Gupta *et al.*, 2017) targeted towards optimizing models' classification performance. However, a model's performance is significantly dependent on how the default or bankruptcy event is defined in the first place. Unfortunately, this aspect has not received sufficient attention in bankruptcy literature. Moreover, legal bankruptcy filings in the United States are becoming an increasingly rare phenomenon¹, and this might provide false signals on firms' default likelihood to external stakeholders. Waiting until the bankruptcy event might lead to significant erosion in firms' value (see Glover, 2016), losses to creditors, and unfavourable business decisions in the years prior to the bankruptcy filing. Thus we also argue for the need for an alternative mechanism to identify firms in financial difficulties, and penalize them for getting close to questionable going-concern status. As a consequence, we propose a

¹ Out of about 7,538 firms in our sample, between 1990 and 2016, only 216 exit the Compustat database due to Chapter 7/11 filings.

definition of financial distress contingent upon firms' earnings, financial expenses, market value and operating cash flow. Our proposed new definition of financial distress essentially builds on that offered by Pindado *et al.* (2008), which is considered to be more functional than its predecessors. We define a firm as financially distressed if it satisfies the three simple conditions: (i) there is negative growth in the average market value (AMV) of the firm in year $t - 1$ and $t - 2$, where AMV is the past twelve months' arithmetic mean of the firm's monthly market values; (ii) its ratio between earnings before interest tax depreciation and amortization (EBITDA) to financial expenses (FE) is less than 1 in the year $t - 1$ and $t - 2$; and (iii) its operating cash flow is less than its financial expenses in the past two years.

Subsequently, we explore the information content of tail risk measures, namely Value-at-risk (VaR) and Expected Shortfall (ES), in explaining firms' likelihood of entering financial distress. Downside or tail risk measures focus only on the risk of underperforming a defined benchmark return, and ignore the risk of outperforming the same. We consider the relationship between downside risk measures and financial distress for several reasons. First, there exists substantial literature pertaining to safety-first investors, whose objective is to minimize the probability of financial distress/bankruptcy. A safety-first investor's portfolio is intended to maximize expected return subject to downside risk constrain (see among others Roy, 1952; Arzac and Bawa, 1977). Second, as part of efficient risk management practices, financial and non-financial firms are increasingly required to quantify the amount of risk their portfolio or assets may incur over specified time horizons. For instance, banks assess their potential losses in order to maintain adequate levels of buffer capital. Similarly, credit rating agencies track the value of firms' assets in order to provide them with an optimum credit rating. Finally, empirical evidence typically shows a skewed distribution of stock returns, with peaks around the mode and fat tails (see among others Jansen and De Vries, 1991, and Conrad *et al.*, 2013). This implies that occurrence of negative extreme events is more frequent than suggested by the

normal distribution. Therefore, traditionally used measures of market risk (i.e. variance or standard deviation) might be insufficient to approximate the likelihood of maximum loss that a firm may witness under highly volatile or normal periods².

There is a growing literature on downside risk and expected stock returns (see for example Bali *et al.*, 2009; Huang *et al.*, 2012). However, we did not come across any suitable literature that explores the relationship between tail risk measures and financial distress. Thus, we explore this relationship, and besides proposing a definition of financial distress, we assess the marginal discriminatory power of tail risk measures, namely VaR and ES, in predicting the financial distress of publicly traded firms in the United States. The relationship we explore is intertemporal as we use three-month, six-month, one-year, three-year and five-year daily returns to measure the VaR and ES of all firms in our sample. Following Zangari (1996), Jaschke (2002), and Liang and Park (2010), we add the Cornish and Fisher (1938) expansion into these risk measures. Contrary to the normal assumption of these risk measures, the Cornish and Fisher (1938) expansion brings skewness and kurtosis into the equation, which is more appropriate for non-normal financial returns. We estimate the risk measure of each firm at the year-end to predict firms' financial distress in the following year. In terms of tail risk measures, our study closely relates to Bali *et al.* (2009), who study the intertemporal relationship between tail risk and expected return, and Liang and Park (2010), who explore the relationship between downside risk measures and hedge funds' failure.

We test our hypothesis using a sample of publicly traded firms in the United States obtained from the Compustat database, covering an analysis period from 1990 to 2016. To gauge the statistical significance of respective tail risk measures in predicting likelihood of financial distress, we begin the empirical validation by reporting univariate regression

²Empirical literature reports that VaR is reasonable in predicting catastrophic financial market events. It also efficiently captures the rate of occurrence of such events, including extreme cases. On the other hand, the traditional measures of market risk, such as standard deviation and conditional variance, provide an inaccurate prediction of extreme events in financial markets (Jorion, 2000; Longin, 2000; Neftci, 2000; Bali, 2003).

estimates for them. Although all tail risk measures and their respective average marginal effects (AME) are highly significant in predicting financial distress likelihood, but VaR measures estimated at the 5% significance level show the highest value of AME for all respective durations. In support of our hypothesis, the univariate regression results established the positive relationship between firms' distress likelihoods and extreme negative returns. We also test the marginal discriminatory power of tail risk measure in the multivariate setup. First, we develop our one-year financial distress prediction model using logistic regression technique and standard financial ratios found significant in recent literature (e.g. Campbell *et al.*, 2008; Gupta *et al.*, 2017). Next, we supplement respective tail risk measures to this baseline distress model, and analyse its significance in the presence of other competing covariates. Empirical results suggest that VaR estimated over longer duration (three and five years) are significant in the multivariate setup and are superior to standard deviation as a proxy for risk measure (as AME of standard deviation is much lower than AME of significant VaR measures). ES is significant for medium and longer duration estimates (one, three and five years respectively), but only measures estimated at the 1% significance level. Considering the theoretical superiority of ES over VaR (see Righi and Ceretta (2016) for additional discussion) we suggest using longer duration ES measures estimated at the 1% significance level as a proxy for risk measure. Finally, we find strong support for our hypothesis that firms that show more frequent extreme negative daily equity returns are more likely to experience the risk of financial distress.

The remainder of the paper is organized as follows: Section 2 defines and provides the rationale behind our proposed definition of financial distress; Section 3 provides definitions of downside risk measures employed in this study; and Section 4 discusses data and financial covariates. Section 5 presents our empirical results, and Section 6 concludes this study.

2. Defining Financial Distress

Traditionally, most academic studies on bankruptcy prediction gravitated around the choice of explanatory variables and modelling methodologies targeted towards improving prediction models' classification performance. However, these performances depend significantly on how the default or bankruptcy event is defined in the first place. Unfortunately, this aspect has not received sufficient attention in the bankruptcy literature. The vast majority of studies employ some legal definition of bankruptcy in line with the relevant bankruptcy code (e.g. Chapter 7/11 filings in the United States), or legal bankruptcy filings supplemented with other distress events, such as delisting from stock exchanges due to financial reasons (e.g. Shumway, 2001), or receiving below investment grade credit ratings (e.g. Campbell et al., 2008). Plausible reasons behind this additional use of proxy bankruptcy events are attributed to the lack of a standard dataset of bankruptcy events, supplemented with the very small frequency of such events. Since these events significantly threaten firms' degree of solvency, these may be used to model firms' default risk. However, these definitions of default may suffer from several noteworthy issues. Firstly, by combining all of these different forms of default, we are trying to predict the probability of a heterogeneous outcome variable, which may be inappropriate. Additionally, since events like bankruptcy resolution involve lengthy legal processes, there often exists a significant time gap between real/economic default date, and legal default date. Moreover, these events are significantly affected by differences in bankruptcy codes between different countries, thus cross comparison of default prediction models between different countries might be inappropriate. Even in cases of uniform codes, instances of judicial discretion cannot be completely ignored where bankruptcy outcomes are often biased toward debtors or creditors (see Gennaioli and Rossi (2010) for additional discussion). Additionally, differences in creditors' rights also play a significant role in bankruptcy or default outcomes. Davydenko and Franks (2008) report large differences in

creditors' rights across France, Germany and the United Kingdom, leading banks to adjust their lending and reorganization practices in an attempt to avoid costly proceedings of bankruptcy filings. This might lead to false impressions about firms' financial health, which might otherwise have questionable going-concern value. A recent study by Franks and Loranth (2014) confirms that the large majority of bankrupt Hungarian firms in their sample were maintained as going-concerns despite these firms having suffered large operating losses, primarily due to varying controlling rights of creditors as per the Hungarian bankruptcy code.

However, if we define financial default as debtors' inability or unwillingness to honour the terms of debt contract (specifically failure to make payment when it is due), then this usually precedes events such as legal bankruptcy filing, delisting, or rating downgrade. Frequent occurrence of financial default events might lead to a state of financial distress, where a firm's going-concern aspect is seriously threatened. Once a firm is financially distressed, it may opt for formal reorganization involving the court system, or an informal reorganization through the market participants (see John *et al.* (2013) for a comparative discussion). Debt restructuring, asset sale, and infusion of new capital from external sources are the three most commonly used market-based or private methods of resolving financial distress (Senbet and Wang, 2010). Debt restructuring allows a financially distressed firm to renegotiate the outstanding debt obligation or related credit terms with its creditors, but is critically subject to whether the debt obligation is due to private or public entity. As an alternative to this, a distressed firm may sell off some of its existing assets to reduce its outstanding liability, or may undertake new profitable investment opportunities, which may eventually help it to overcome its misery. Despite having profitable investment opportunities, a financially distressed firm might not be able to generate additional funding, due to the high risk involved in financing distressed firms and the "debt overhang" problem as discussed by Myers (1977). As a consequence, infusion of new capital from external sources is rarely observed in the

resolution of financial distress. Thus, we cannot rule out the possibility that a financially distressed firm may not file for Chapter 7 or Chapter 11 protection and choose a private workout method for resolving financial distress. Gilson *et al.* (1990) and Gilson (1997) report that firms avoid legal bankruptcy processes by out-of-court negotiation with creditors. However, under the binary classification based on legal consequences, a financially distressed firm which has not filed for Chapter 7 or Chapter 11 is not considered to have questionable going-concern value. There is therefore a clear need for a mechanism to identify financially distressed firms beyond the legal definitions. Waiting until the bankruptcy event might lead to significant erosion in firms' value, losses to creditors, and unfavourable business decisions in the years prior to the bankruptcy filing year. It is in the interest of most stakeholders to raise the flag when the symptoms of financial distress are visible, in order to avoid or prepare for the forthcoming crisis. In this regard, lending or credit decisions based on firms' *financial distress* likelihood may be more relevant in addressing respective stakeholders' objectives.

Moreover, from literature pertaining to bankruptcy costs we understand that direct costs of bankruptcy, including legal and liquidation expenses, are usually followed by indirect costs, which are generally in the form of opportunity costs leading to declining degrees of solvency. This is in line with the theoretical arguments of Purnanandam (2008), who suggests that financial distress is a state in which firms are unable to meet their debt obligations, ultimately leading them to insolvency as the debt matures. Prominent signs of indirect costs include loss of valuable employees, restrictive terms of trade credit, and decline in consumer confidence and investment growth, which over time may lead to significant loss in market value even if the firm never files for legal bankruptcy (Chen and Merville, 1999). Thus a financially distressed firm is left with the challenge of persuading employees, customers, suppliers and trade creditors to engage in business with it. The closer to bankruptcy a firm is, the stronger this challenge becomes. Stakeholders begin abandoning distressed firms, leading to even faster

depletion of operating profits and shareholder wealth. Thus rather than considering a firm bankrupt when it files for legal bankruptcy, it is appropriate to penalize it for being sufficiently close to bankruptcy/default for their going-concern status to become questionable.

Considering the discussion above we believe that financial distress is more practically relevant than legal bankruptcy events, and in the following discussion we propose a new definition of financial distress that builds on suggestions and findings from earlier related studies. In this study, we define a firm as financially distressed in year t if it satisfies the following three conditions:

Condition 1: There is negative growth in the average market value (AMV) of the firm in year $t - 1$ and $t - 2$. Here, AMV is the past twelve months' arithmetic mean of the firm's monthly market values.

Although decline in firms' market value in two consecutive time periods has been suggested in past literature (e.g. Pindado et al., 2008), we suggest considering the average market value rather than the end of the analysis year market value, as the latter represents a static snapshot of a firm's market capitalization at the end of a given trading day, which may be significantly influenced by negative or positive developments around that trading day. Thus, instead of market value on a given day, we use AMV, as it is a better representation of the average state of a firm.

Condition 2: The ratio between earnings before interest tax depreciation and amortization (EBITDA) to financial expenses (FE) is less than 1 in the year $t - 1$ and $t - 2$.

Traditionally, excessive leverage was believed to be the primary cause of bankruptcy or financial difficulties (e.g. John *et al.*, 2013), however significant volumes of literature argue that firms with a higher cost of default choose a lower level of leverage to mitigate potential

default likelihood (e.g. George and Hwang, 2010; Glover, 2016), and thus we cannot rule out the possibility of financial distress triggered by reasons other than leverage. However, failure of such firms should not bring significant worries to providers of debt capital, thus by including a condition on firms' ability to meet their financial expenses, we are essentially trying to capture those firms that enter into financial difficulties due to their inability to meet financial commitments to the providers of debt capital. Pindado *et al.* (2008) suggest that financially distressed firms should report financial expenses higher than their earnings in two consecutive years. This effectively implies that the ratio between earnings to financial expenses should be less than one in these two time periods.

Condition 3: the firm's operating cash flow (OCF) is less than its financial expenses in the year $t - 1$ and $t - 2$.

If we define financial default as debtors' inability to make payment when it is due, then cash flow is of prime importance. Since the adoption of an accrual accounting system for recording and reporting business transactions, balance sheets and income statements have emerged as dominant sources of information for business decision-making. However, accruals are the non-cash component of earnings, and represent adjustments made to cash flows to generate a profit measure largely unaffected by the timing of receipts and payments of cash (Ball *et al.*, 2016). Thus, just having earnings above financial expenses does not guarantee a firm's ability to repay its debt on time. Inefficient management of cash may also result in premature closure of even profitable businesses, as loans, most operating expenses, taxes etc are required to be paid in cash. However, this important cash flow information is often overlooked in standard financial analysis and related decision-making (e.g. Pindado *et al.*, 2008). Ball *et al.* (2016) argue that cash-based measures of operating profitability (a measure that excludes accruals) outperform measures of profitability that include accruals. Additionally,

Grullon *et al.* (2017) report that, for large firms, cash flow remains the primary driver of investment spending, and the marginal effect of cash flow on aggregate investment has actually increased over the past 30 years. Recent literature also highlights the importance of financing constraints and free-cash-flow problems in making sound investment decisions (Lewellen and Lewellen, 2016). The relative importance of cash flow in times of financial distress is also realized in recent literature. Lee *et al.* (2016) argue that, in times of financial distress, investors place significantly higher weight on OCF information than on earnings information, as they find that a firm's stock returns show stronger association with its OCF than its earnings when the firm is in financial distress. Considering the discussion above, and to be consistent with the two previous conditions, we include *Condition 3* in our proposed definition of financial distress, with an aim to overcome some of the limitations of firms' failure definitions discussed earlier.

3. Estimating Downside Risk Measures

We use three-month, six-month, one-year, three-year and five-year daily returns to measure VaR and ES of all firms in our sample. We estimate the risk measure of each firm as of the year-end to predict firms' financial distress in the next year. For instance, to predict a firm's financial distress in the year 2005, we calculate its risk measure as of December 2004. Thus, the actual data we use for estimating three-month VaR is from the period October 2004 to December 2004. We apply a similar rule to estimate other risk measures.

Value-at-risk: Estimation of VaR requires the definition of two basic parameters, the confidence level ($1 - \alpha$) and the time horizon (τ), along with the estimation model. These are then used to estimate the worst loss that may occur over the specified time horizon and confidence level.

Traditionally, the risk horizon is measured in the number of trading days³, rather than the number of calendar days for measuring VaR, and it is this period over which we measure the expected loss. In the absence of any defined rules or guidelines, time horizon is usually chosen by considering the liquidity of the risk and the duration of exposure to that risk (Alexander, 2008). The higher the assets' liquidity, the lower the time required to close or fully hedge the exposure. Thus, the time horizon required is shorter for the VaR model. A longer time horizon is suggested when measuring VaR in stressful and volatile market conditions, as markets tend to lose liquidity during such economic conditions. Credit rating agencies assign top ratings to those companies that exhibit extremely low default probability (such as 0.03%) over the next one-year horizon. Hence companies looking for AA or above credit rating should apply a confidence level of 99.97% to measure their enterprise-wide VaR over the next one-year horizon. Considering the arguments presented above, and the nature of our application, we estimate downside risk measures using five different time horizons to detect the presence of any intertemporal differences. We estimate the risk measure of each firm as of the year-end using the past three-month, six-month, twelve-month, three-years and five-years daily returns, to predict firms' financial distress the following year.

The choice of confidence level is also highly subjective; it primarily depends on the area of application and users' attitude towards risk. For instance, the Basel II Accord requires banks using internal VaR models to assess their capital requirements to measure VaR at a 99% confidence level, while a credit rating agency may maintain a more stringent confidence level of about 99.97% to assess the probability of firms' insolvency hazard (Jorion, 2000). On the other hand, a risk manager may allow a lower confidence level (95%) and shorter risk horizon (1 day) while setting VaR based trading limits. Thus we consider two different loss probability

³ In this study, our risk estimates are computed using 252 trading days in a year.

levels, namely 99% ($\alpha = 0.01$) and 95% ($\alpha = 0.05$) confidence levels, to estimate the downside risk measures.

Let τ denote the time horizon, $R_{t+\tau}$ denote a firm's return between the time period t and $t + \tau$, and let $F_{R,t}$ represent the cumulative distribution function (CDF) of $R_{t+\tau}$ conditional upon the set of information available at time t . Then, $F_{R,t}^{-1}$ represents the inverse function of $F_{R,t}$. Given this, the VaR of a firm's return as of time t with a time horizon τ and at $(1 - \alpha)$ confidence level can be estimated as follows:

$$VaR_t(\alpha, \tau) = -F_{R,t}^{-1}(\alpha) \quad (1)$$

Unlike normal VaR, the semi-parametric Cornish-Fisher VaR (VaR_{CF}) considers higher moments in the return distribution, thus relaxing the normality assumption. The Cornish and Fisher (1938) approximation incorporates higher moments (skewness and kurtosis) into the estimation process and thus accounts for non-normality in the return distribution. Zangari (1996) is the first to introduce this in his study that estimates the VaR of option portfolios. The justification behind the use of this formula is that any distribution can be approximated with these known moments, in terms of any other given distribution (Johnson *et al.*, 1994; Jaschke, 2002). Thus, we use VaR_{CF} ⁴ to accommodate non-symmetrical and fat-tailed returns distribution. The fourth order Cornish and Fisher (1938) expansion for α percentile of $(R - \mu)/\sigma$ is shown in equation (2), while equation (3) defines the Cornish-Fisher VaR (VaR_{CF}).

$$\Omega(\alpha) = Z(\alpha) + \frac{1}{6}(Z(\alpha)^2 - 1)S + \frac{1}{24}(Z(\alpha)^3 - 3Z(\alpha))K - \frac{1}{36}(2Z(\alpha)^3 - 5Z(\alpha))S^2 \quad (2)$$

$$VaR_{CF} = -(\mu + \Omega(\alpha) \times \sigma) \quad (3)$$

⁴ We use VaR and VaR_{CF} interchangeably in this manuscript.

where, σ is the standard deviation, μ is the average return, S is the measure of skewness, K is the excess kurtosis of past n -month daily returns, $(1 - \alpha)$ is the confidence level, and $Z(\alpha)$ is the critical value obtained from the standardized normal distribution⁵.

Expected Shortfall: Although VaR is the maximum loss that can take place over a defined time horizon and confidence level, there still exists a very small probability that a loss greater than VaR may be experienced. Thus, VaR does not reveal any information about the size of the loss in the event when that level is breached. Expected Shortfall (ES) provides this information. It is the conditional expected loss that is greater than or equal to the VaR. ES expressed in terms of return rather than dollar amount is formulated as follows:

$$\begin{aligned}
 ES_t(\alpha, \tau) &= -E_t[R_{t+\tau} | R_{t+\tau} \leq -VaR_t(\alpha, \tau)] \\
 &= -\frac{\int_{v=-\infty}^{-VaR_t(\alpha, \tau)} v f_{R,t}(v) dv}{F_{R,t}[-VaR_t(\alpha, \tau)]} \\
 &= -\frac{\int_{v=-\infty}^{-VaR_t(\alpha, \tau)} v f_{R,t}(v) dv}{\alpha}
 \end{aligned} \tag{4}$$

Here, $R_{t+\tau}$ represents a firm's return in the time period t , and $t + \tau$ and $f_{R,t}$ denotes the conditional probability density function (PDF) of $R_{t+\tau}$. $F_{R,t}$ is the conditional CDF of $R_{t+\tau}$ conditional upon the information set available at time t , $F_{R,t}^{-1}$ represents the inverse function of $F_{R,t}$, and $(1 - \alpha)$ is the confidence level. Artzner *et al.* (1999) provide the theoretical argument that VaR is inferior to ES as a risk measure. They argue that, unlike VaR, ES possesses mathematical properties like continuity and subadditivity which are desirable to a coherent

⁵Note that standard deviation is always positive, while the original VaR and ES are usually negative. To avoid confusion, the original VaR and ES numbers are multiplied by -1 in equations (3) and (4). Therefore, the VaR and ES numbers presented in this paper are usually positive.

measure of risk. We use Cornish-Fisher VaR (Var_{CF}) calculated using equations (2) and (3) to estimate Expected Shortfall, denoted as ES_{CF} ⁶.

4. Data and Covariates

This section presents discussion on our sources of data and choice of covariates used in this study to perform required empirical analysis.

4.1 Data

Unlike several others (e.g. Chava and Jarrow, 2004), we do not try to predict the financial distress event over the next month or quarter. We believe that an efficient distress prediction model should work as an Early Warning System (Edison, 2003) and should signal any adverse distress event well in advance. This may provide sufficient time to avert or manage the forthcoming crisis. Thus, to predict the financial distress over the next one-year horizon, our empirical analysis employs annual firm-level accounting and market data of the United States firms from the Compustat database. We consider a relatively long analysis period from 1990 to 2016. We also limit our sample to firms which are publicly traded either on AMEX, NASDAQ or NYSE stock exchanges only. Transportation, Communications & Public Utilities; Finance, Insurance & Real Estate; and Public Administration firms have been excluded from our analysis (see Table 1).

[Insert Table 1 Here]

4.2 Covariates

Dependent Variable: We use the definition of financial distress proposed in Section 2 as the binary dependent variable for this study.

⁶ We use ES and ES_{CF} interchangeably in this manuscript.

Independent Variables: A large volume of empirical literature has evolved over time which efficiently addresses the issues pertaining to prediction of corporate bankruptcy. However, the default prediction models gravitate toward the use of accounting variables (e.g. Altman, 1968; Ohlson, 1980) or market information to predict firms' insolvency hazard (e.g. Bharath and Shumway, 2008). The market-based approach of modelling default risk has become popular since their commercial introduction by KMV, CreditMetrics and others. Hillegeist *et al.* (2004) report that market-based structural models perform better than popular accounting-based models, in particular the Altman Z-score (Altman, 1968) and Ohlson O-score (Ohlson, 1980) models. In a similar study, Das *et al.* (2009) report that their distress prediction models, built using accounting metrics and market metrics respectively, exhibit very similar classification performances. Models which employ both accounting and market information substantially outperform both accounting and market based models, suggesting that accounting and market information are complementary to each other. Findings of other related studies (see among others Agarwal and Taffler, 2008; Trujillo-Ponce *et al.*, 2014) also support the findings of Das *et al.* (2009). Considering the above arguments, we use accounting and market information to develop our distress prediction models. In particular, we employ all eight (three accounting-based and five market-based) measures of bankruptcy employed by Campbell *et al.* (2008) to predict the bankruptcy of US firms, as they report that this set of covariates outperforms other standard sets of covariates employed in prior studies. We supplement this with two additional covariates, financial expenses to sales (FES) and tax to market valued total assets (TMTA), as they exhibit significant explanatory power and very high average marginal effects in several recent studies on SMEs financial distress (e.g. Gupta *et al.* 2017). Furthermore, we also test the marginal predictive ability of tail risk measures discussed in Section 3, i.e. VaR and ES. Detailed definitions of financial covariates are as follows:

- I. Total liabilities to Market-valued Total Assets (TLMTA), calculated as:

$$TLMTA_{i,t} = \frac{Total\ Liabilities_{i,t}}{(Market\ Value\ of\ Equity_{i,t} + Total\ Liabilities_{i,t})}$$

Here a Market-valued Total Asset (MTA) is the sum of the market value of a firm's equity and its total liabilities. In case of missing values of Total Liabilities, MTA is equal to the firm's equity, and vice-versa.

- II. Cash and short term assets to Market-valued Total Assets (CMTA), calculated as:

$$CMTA_{i,t} = \frac{Cash\ and\ Short\ Term\ Investments_{i,t}}{(Market\ Value\ of\ Equity_{i,t} + Total\ Liabilities_{i,t})}$$

- III. Weighted average of Net Income to Market-valued Total Assets (*NIMTAAVG*) over the previous 3 year period (*NIMTAAVG*), calculated as:

$$NIMTAAVG_{t-1,t-3} = \frac{1}{1.75} NIMTA_{t-1} + \frac{0.5}{1.75} NIMTA_{t-2} + \frac{0.25}{1.75} NIMTA_{t-3}$$

where

$$NIMTA_{i,t} = \frac{Net\ Income_{i,t}}{(Market\ Value\ of\ Equity_{i,t} + Total\ Liabilities_{i,t})}$$

Following Chen and Hill (2013), we modify Campbell *et al.* (2008)'s measure of average *NIMTA* to allow for the fact that we employ annual, rather than quarterly, accounting measures. Missing values of *ANIMTA* are replaced with the cross-sectional mean of *NIMTA*.

- IV. Weighted average of monthly log excess return relative to value-weighted S&P 500 return over the previous 12 months period (*EXRETAVG*), calculated as:

$$EXRETAVG_{t-1,t-12} = \frac{1 - \phi}{1 - \phi^{12}} (MEXRET_{t-1} + \dots + \phi^{11} MEXRET_{t-12})$$

Where;

$$MEXRET_{i,t} = \log(1 + Return_{i,t}) - \log(1 + Return_{S\&P\ 500,t})$$

Missing values of AMEXRET are replaced with the cross-sectional mean of MEXRET.

- V. Annualized standard deviation of a firm's daily log returns over the previous three months (SIGMA), estimated as:

$$SIGMA_{i,m-1,m-3} = \left(252 \times \frac{1}{N-1} \sum_{k \in \{m-1, m-2, m-3\}} r_{i,k}^2 \right)^{\frac{1}{2}},$$

where $r_{i,k}^2$ represents the log returns of firm i on the day k , and N represents the number of days in the three-month estimation window. Here, following Campbell *et al.* (2008), we use a proxy centred around zero rather than three months' rolling mean.

- VI. Firm's Market-to-book ratio (MB)
- VII. Log of price per share winsorized above \$15 (PRICE)
- VIII. Logarithm of each firm's size relative to S&P 500 market capitalization (RSIZE), calculated as:

$$RSIZE_{i,t} = \log \left(\frac{Firm\ Market\ Equity_{i,t}}{Total\ S\&P\ Market\ Value_t} \right)$$

- IX. Financial Expense to Sale (FES), calculated as:

$$FES_{i,t} = \frac{Financial\ Expense_{i,t}}{(Sale_{i,t})}$$

- X. Tax to Market-Valued Total Assets (TMTA), calculate as:

$$TMTA_{i,t} = \frac{Total\ Income\ Tax_{i,t}}{(Market\ Value\ of\ Equity_{i,t} + Total\ Liabilities_{i,t})}$$

Healthy firms are expected to pay higher taxes due to higher business volume in comparison to distressed firms, and hence, a negative relationship is expected between TMTA and financial distress likelihood. Further, following the suggestion of Cohen *et al.* (2003), we adjust the book value of equity to avoid the influence of outliers. Specifically, we add 10% of the difference

between the market value and the book value of equity to the book value of equity⁷, thus increasing book values that are relatively very small and probably wrongly measured. This ultimately prevents outliers when book equity is used in the denominator when calculating financial ratios. For observations which still exhibit negative book value of equity after this adjustment, we replace negative book values with positive \$1 to ensure that market-to-book value ratios for all observations are positive. Further, we restrict the range of all the covariates within 5th and 95th percentiles to limit the influence of any outliers.

Control Variables: To control the volatility in the macroeconomic environment affecting specific industrial sectors, we calculate a measure of industry risk (*INDRISK*) as the financial distress rate (number of firms experiencing the event of interest in the respective industrial sector in a given year/total number of firms in that industrial sector in that year) in each of the seven industrial sectors in a given year. Higher values indicate a higher risk of financial distress, and vice versa.

[Insert Table 2 Here]

5. Empirical Methods

5.1 Logistic Regression

Although we see a significant rise in the popularity of hazard models in modelling bankruptcy or financial distress events, in line with the findings of Gupta *et al.* (2017) we use panel logistic regression to establish our empirical validation. Gupta *et al.* (2017) argue that the discrete-time hazard model with logit link is essentially a panel logistic model that controls

⁷ $Total\ Assets(adjusted)_{i,t} = Total\ Assets_{i,t} + 0.1(Market\ Equity_{i,t} - Book\ Equity_{i,t})$

for firms' age. Thus we assume that marginal probability of firms' financial distress over the next year follows a logistic distribution that is estimated as follows:

$$P(Y_{it} = 1) = \frac{1}{1 + \exp(-\alpha - \beta X_{i,t-1})} \quad (7)$$

where Y_{it} is an indicator variable that equals one if the firm is in financial distress in year t , and $X_{i,t-1}$ is a vector of explanatory variables known at the end of the previous year. To capture any duration dependency, we use the natural logarithm of firms' annual age⁸ (AGE) as a control variable in our multivariate models.

5.1 Summary Statistics and Correlations

Table 3 provides an overview of mean, median, standard deviation, minimum and maximum of all covariates used in our study. The first column gives a list of all covariates used, and the second column shows which group these covariates belong to. '0' represents the non-distressed/censored group, '1' represents the distressed group, and 'All' is the entire dataset. This gives an initial understanding about the variability of our covariates. Descriptive measures of the vast majority of covariates are as per our expectation and without any extreme variability.

The mean/median of covariates we report are very similar to those reported by Campbell *et al.* (2008) in their study, with some differences due to the data period and different definition of the default event. We expect the mean/median of the covariates that are positively related to the insolvency/distress risk to be higher for the distressed group than for the group without distress (see for example, variable FES in Table 3). Contrarily, we expect the mean/median of the covariates that are negatively related to the insolvency/distress risk to be lower for the

⁸ We proxy a firm's age as the earliest year for which financial information for that firm is available in the Compustat database.

distressed groups than for the group without distress (see for example, variable NIMTAAVG in Table 3). However, we also see some exceptions. The mean value of TLMTA for the distressed group is lower than for the censored group of firms, implying that financially distressed firms have lower total liabilities. This is contrary to our traditional understanding, where we expect firms in financial distress to have higher liabilities compared to their total assets. Similarly, the mean value of CMTA for the distressed group is higher than for the censored group, implying that financially distressed firms have higher cash reserves. This is contrary to intuition, where we expect financially distressed firms to have liquidity problems, and hence lower reserves of cash and short-term investments. Also, the mean market-to-book (MB) ratio, which is expected to be lower for the distressed group of firms than its censored counterpart, is actually higher. This may suggest active demand of distressed stocks, which consequently push their stock price higher. These contrary revelations might affect the sign of respective regression coefficients.

[Insert Table 3 Here]

A comparison of the distressed and non-distressed groups also reveals expected differences in Table 3. For example, distressed firms make mean loss of about 18%, compared with non-distressed firms that make mean loss of only about 1.9%, as lower mean of net income relative to market value of total assets shown by NIMTAAVG. The mean volatility of distressed firms is quite high at 82%, relative to only 55% of non-distressed firms, as shown by CSIGMA. Distressed firms are relatively small (RSIZE), and have a much lower mean log price per share (PRICE), and lower returns (EXRETAVG) compared to their non-distressed counterpart. Similar to Campbell *et al.* (2008), the mean of market-to-book (MB) ratio of financially distressed firms is slightly higher, because the book value of their equity erodes due

to frequent losses. This does not seem to be reflected in the market value of the firms because of the proactive nature of stock markets.

A closer look at extreme measures generated using three-month, six-month, one-year, three-year and five-year daily returns (i.e. VaR and ES) reveals that distressed firms have higher values compared to non-distressed firms. This is to be expected, as distressed firms tend to exhibit extreme movements. The mean/median of downside measures with the 95% confidence interval are lower compared to the measures with the 99% confidence interval. This is also intuitive as, at the lower confidence level, the cut-off point at which the loss should not exceed should be lower in case of VaR, and at the higher confidence level, the cut-off point should be higher. Similarly, the mean of the losses larger than VaR (i.e., ES) should be lower at lower confidence intervals and higher at higher confidence intervals. We also find intertemporal differences, especially at the 99% confidence interval. We see the mean/median of extreme measures at shorter rolling windows are less than those at longer rolling windows. However, at the 95% confidence interval, the mean/media of extreme measures is slightly higher.

Most of the covariates shown in Table 4 show low to moderate correlation. However, PRICE shows strong positive correlation with RSIZE (0.718) and strong negative correlation with CSIGMA (-0.609). Thus, we take into account the issues of multicollinearity while developing our multivariate models.

[Insert Table 4 Here]

5.2 Univariate Regression and Average Marginal Effects

Table 5 reports the univariate regression results of respective covariates. This table reports point estimates (β), standard errors (SE), z-statistics (Z) and average marginal effects

(AME) in percentages⁹. All financial ratios are highly significant at the 1% level of significance. MB negatively predicts the financial distress of firms, meaning that the stock market is already anticipating the financial distress, driving down the market value of equity and hence the market-to-book ratio. As explained above in the summary statistics, we find another counter intuitive result as CMTA positively predicts the likelihood of financial distress.

Seven out of twelve covariates show AME of less than 5% with AME of TLMTA, MB, RSIZE and AGE even less than 1%. However, the other five covariates reported are highly significant predictors, with more than 5% AME. Two non-Campbell covariates, FES and TMTA, exhibit the highest AME at around 5% and 30% respectively. Among extreme risk measures, all are highly significant at the 1% level of significance. This implies that all extreme risk measures can predict financial distress, but some of the measures have very low values of AME that raise scepticism about their predictive power in the multivariate setup. We see intertemporal differences as longer horizon rolling estimates show more statistical and economic significance. For example, three-year and five-year VaR, estimated at the 5% level of significance, have the highest AME of 5.93% and 7.15% respectively. Although the AME of ES are not as high as for the VaR, three-year and five-year ES estimated at the 1% level of significance have the highest AME of 1.55% and 1.59% respectively. This shows that its discriminatory power is sensitive to duration and the significance level of tail risk estimates.

[Insert Table 5 Here]

5.3 Baseline Multivariate Model

In order to test the marginal discriminatory power of downside risk measures, we first estimate the baseline multivariate model with all financial covariates, as they are all significant

⁹ AME is the change in the outcome probabilities due to unit change in the value of a given covariate, provided that other covariates are held constant.

in the univariate analysis. However, considering that few cases of moderate to high correlation among covariates are reported in Table 4, we follow the method suggested by Gupta *et al.* (2017) while developing our baseline multivariate model. Primarily, this requires the ranking of covariates in univariate regression based on the magnitude of their respective AME, and then introducing each variable at a time in the multivariate setup, in declining order of their AME. Gupta *et al.* (2017) argue that the higher the value of AME, the higher the change in the predicted probability due to unit changes in the covariate's value. Also, we exclude a covariate from the multivariate model if, when introduced: (i) it reverses the expected sign of any previously added covariate; (ii) it bears the opposite sign to that expected; (iii) it bears the expected sign but has a p-value greater than 0.10; and (iv) it makes a previously added covariate insignificant with a p-value greater than 0.10. This allows us to develop a baseline multivariate model that excludes RSIZE and MB. Additionally, RSIZE also exhibits strong correlation with PRICE and moderate correlation with SIGMA. We also exclude SIGMA from our baseline model, as the purpose of this study is to report the comparative performance of several tail risk measures in comparison to SIGMA, which we do in the subsequent section. Finally, we include INDRISK and natural logarithm of firms' annual age (AGE) as control variables in the model. As reported in Table 6, we develop the final baseline model with nine covariates, all of which are highly significant and retain their respective expected signs. We therefore conclude that all of these covariates are jointly significant in predicting the financial distress of US firms. Also, as reported in Figure 1, the within-sample and out-of-sample area under ROC curves (AUROC¹⁰) of this model is about 93%, suggesting excellent performance of our proposed definition of financial distress.

¹⁰ The higher the AUROC, the better the prediction model. Please see Gupta *et al.* (2016) for additional discussion on this. For out-of-sample validation, we first estimate models using observations until the year 2011, and using these estimates we predict financial distress probabilities for the year 2012; we then include 2012 in the estimation sample and predict distress probabilities for 2013 and so on, until the year 2016. We then use these predicted probabilities from the year 2012 through 2016 to estimate out-of-sample AUROC.

[Insert Table 6 Here]

[Insert Figure 1 Here]

5.4 Multivariate Models with Risk Measures

In order to test the marginal discriminatory power of tail risk measures in comparison to SIGMA, we re-estimate the baseline multivariate model (as reported in Table 7), supplemented with tail risk measures reported in Table 2. Rows 3 to 7 in Table 7 and Table 8 report regression estimates for respective risk measures as listed in the first row of columns 2 to 12. $LR\ Chi^2$, in the seventh row of column (1), is the chi-square for likelihood ratio test that compares the log likelihood of the baseline model supplemented with risk variables with the log likelihood of the baseline model, and tests whether this difference is statistically significant. To capture any variations that may arise due to the difference in significance levels of tail risk estimates, we estimate separate models employing tail risk measures estimated at 1% and 5% significance levels. Furthermore, we estimate separate multivariate models for three-month, six-month, one-year, three-year, and five-year intervals, to account for any intertemporal differences that may exist. Following Liang and Park (2010), we compare these models with our baseline multivariate model, supplemented with SIGMA as the risk measure. Due to the moderate correlation between PRICE and all risk measures (see Table 4), signs of respective risk measures turning insignificant or opposite are expected. Thus we exclude PRICE from this analysis as our primary objective is to assess the marginal discriminatory power of respective risk measures.

We estimate all multivariate models using our proposed definition of financial distress as defined in Section 2. Table 7 reports our multivariate regression results with VaR measures. The first column gives the names of the covariates. Column (2) reports estimates with SIGMA as the risk measure (without PRICE). Columns (3) to (7) report respective regression estimates

employing three-month, six-month, one-year, three-year and five-year estimates of tail risk measured at the 99% confidence interval. Columns (8) to (12) report the same but for tail risk measures, estimated using the 95% confidence interval.

Table 7 reveals intertemporal differences and that short term rolling estimates of VaR measures are not significant in predicting firms' likelihood of financial distress. As we measure the extreme downside risk, it might be that short term rolling estimates do not have enough observations (Huang *et al.*, 2012),¹¹ therefore longer term rolling estimates perform better in predicting the financial distress of firms. The multivariate model with five-year VaR measured at the 95% confidence interval (VAR5Y5) is the best, as it shows the highest value of AME (5.85%) in comparison to other significant risk measures, and the highest value of $LR\ Chi^2$. The baseline model with SIGMA as a measure of risk is also statistically significant with $LR\ Chi^2$ of 9.15; however, the AME is very low at 0.43%. Other significant downside risk measures are five-year VaR measured at the 99% confidence level, and three-year VaR measured at the 95% confidence level.

Subsequently, we repeat this multivariate analysis with ES as the risk measure and report the results in Table 8. Similar to multivariate regression results with VaR, we find intertemporal differences in that the longer term rolling estimates of ES perform better in predicting the financial distress of firms. However, with ES we find that five-year ES measured at the 99% confidence interval performs better (with the $LR\ Chi^2$ of 19.53) than the five-year ES measured at the 95% confidence interval. Since ES is the mean of the losses greater than VaR, the mean of the losses is lower at the 95% confidence interval because lower values reduce the mean of the losses. The mean is higher at the 99% confidence interval and, therefore, we find stronger results for five-year ES measured at the 99% confidence interval. However, VaR is simply a

¹¹ While measuring ES and TR, Bali *et al.* (2009) also use a longer rolling window of 100 days compared to shorter windows for measuring VaR.

cut-off point, and the cut-off point at the 99% confidence interval covers the cut-off point at the 95% confidence interval. Therefore, we find stronger results for five-year VaR measured at the 95% confidence interval. Other ES measures that significantly predict the financial distress likelihood of firms are one-year and three-year ES measures at the 99% confidence interval. Our baseline model with SIGMA as a measure of risk also significantly explains the financial distress of firms but shows lower values if AME and $LR\ Chi^2$ than ES5Y1. The AME of ES1Y1 and ES3Y1 are also lower than the AME of ES5Y1. Overall, as the superiority is ES over VaR is well documented in the literature, we conclude that five-year ES measured at the 99% confidence interval (ES5Y1) is superior to SIGMA (higher AME than SIGMA) and other competing tail risk measures.

[Insert Tables 7 and 8 Here]

6. Conclusion

Appropriate pricing of credit risk is integral to maintaining sustainable and competitive lending practices. This also informs the estimation of capital reserves that lending institutions are required to maintain in proportion to their risk-weighted assets. Overestimation of credit risk shall lead to higher capital reserves, and thus an opportunity loss, whereas underestimation results in lower capital reserves, but may trigger large shortfalls and significant capital losses (see Demirguc-Kunt *et al.*, 2013, for brief discussion). We contribute to this strand of literature by documenting the impact of tail risk measures on firms' likelihood of financial distress.

At first, we argue the need for a mechanism to identify firms in financial difficulties rather than waiting until they file for bankruptcy. Waiting until the bankruptcy filing might lead to significant erosion in firms' value, losses to creditors and unfavourable business decisions. Thus, there is a need for an alternative mechanism to identify firms in financial

difficulties, and it shall be appropriate to penalize them for getting close to questionable going-concern status. As a consequence, we propose a definition of financial distress contingent upon firms' earnings, financial expenses, market value and operating cash flow. Specifically, we define a firm as financially distressed if it satisfies three simple conditions: there is negative growth in the average market value (AMV) of the firm in year $t - 1$ and $t - 2$; the ratio between earnings before interest tax depreciation and amortization (EBITDA) to financial expenses (FE) is less than 1 in the year $t - 1$ and $t - 2$; and its operating cash flow is less than financial expenses in the past two years. This definition will also be helpful to traders of distressed stock in identifying financially distressed firms.

Next, we hypothesize that more frequent extreme negative daily equity returns result in higher tail risk, and this subsequently increases firms' likelihood of entering financial distress. Thus, we assess the marginal discriminatory power of tail risk measures, namely value-at-risk and expected shortfall, in predicting financial distress likelihood of United States publicly traded firms. Empirical results suggest that longer duration VaR and ES exhibit significant discriminatory powers, however for ES, we observe significance only for 1% significance level estimates.

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Tables & Figures

Table 1: Sample Industrial Classification

Industry Code	SIC Code	Industry	Included/Excluded
1	< 1000	Agriculture, Forestry, Fishing	Included
2	1000 to < 1500	Mining	Included
3	1500 to < 1800	Construction	Included
4	2000 to < 4000	Manufacturing	Included
5	5000 to < 5200	Wholesale Trade	Included
6	5200 to < 6000	Retail Trade	Included
7	7000 to < 8900	Services	Included
Excluded	4000 to < 5000	Transportation, Communications & Public Utilities	Excluded
Excluded	6000 to < 6800	Finance, Insurance & Real Estate	Excluded
Excluded	9100 to < 10000	Public Administration	Excluded

Notes: This table reports Standard Industrial Classification (SIC) of the United States firms. SIC Code is a four digit code that represents a given industrial sector. The last column reports the industrial sectors that we included or excluded from our sample.

Table 2: Variable Description

Variable	Definition	Compustat Data Item
TLMTA	Total liabilities to Market-valued Total Assets	LT, PRCC_F, CSHO
CMTA	Cash and short term assets to Market-valued Total Assets	CHE, PRCC_F, CSHO, LT
NIMTAAVG	Weighted average of Net Income to Market-valued Total Assets over the previous 3 year period	NI, PRCC_F, CSHO, LT
EXRETAVG	Weighted average of monthly log excess return relative to value-weighted S&P 500 return over the previous 12 months period	PRCCM, AJEXM, SPTRN
SIGMA	Annualized standard deviation of a firm's daily log returns over the previous three months	PRCCD, AJEXDI
MB	Firm's Market-to-book ratio	PRCC_F, CSHO, SEQ
PRICE	Log of price per share winsorized above \$15	PRCC_F
RSIZE	Logarithm of each firm's size relative to S&P 500 market capitalization	PRCC_F, CSHO, TOTVAL
FES	Financial Expense to Sale	IE, SALE
TMTA	Tax to Market-Valued Total Assets	TXT, PRCC_F, CSHO, LT
AGE	Natural logarithm of firms' annual age	
INDRISK	Number of firms experiencing financial distress in respective industrial sectors in a given year/total number of firms in that industrial sector in that year	
VAR3M1	Value-at-Risk estimated over past 3 months at 1% significance level	PRCCD, AJEXDI
VAR6M1	Value-at-Risk estimated over past 6 months at 1% significance level	PRCCD, AJEXDI
VAR1Y1	Value-at-Risk estimated over past 1 year at 1% significance level	PRCCD, AJEXDI
VAR3Y1	Value-at-Risk estimated over past 3 years at 1% significance level	PRCCD, AJEXDI
VAR5Y1	Value-at-Risk estimated over past 5 years at 1% significance level	PRCCD, AJEXDI
VAR3M5	Value-at-Risk estimated over past 3 months at 5% significance level	PRCCD, AJEXDI
VAR6M5	Value-at-Risk estimated over past 6 months at 5% significance level	PRCCD, AJEXDI
VAR1Y5	Value-at-Risk estimated over past 1 year at 5% significance level	PRCCD, AJEXDI
VAR3Y5	Value-at-Risk estimated over past 3 years at 5% significance level	PRCCD, AJEXDI
VAR5Y5	Value-at-Risk estimated over past 5 years at 5% significance level	PRCCD, AJEXDI
ES3M1	Expected Shortfall estimated over past 3 months at 1% significance level	PRCCD, AJEXDI
ES6M1	Expected Shortfall estimated over past 6 months at 1% significance level	PRCCD, AJEXDI
ES1Y1	Expected Shortfall estimated over past 1 year at 1% significance level	PRCCD, AJEXDI
ES3Y1	Expected Shortfall estimated over past 3 years at 1% significance level	PRCCD, AJEXDI
ES5Y1	Expected Shortfall estimated over past 5 years at 1% significance level	PRCCD, AJEXDI
ES3M5	Expected Shortfall estimated over past 3 months at 5% significance level	PRCCD, AJEXDI
ES6M5	Expected Shortfall estimated over past 6 months at 5% significance level	PRCCD, AJEXDI
ES1Y5	Expected Shortfall estimated over past 1 year at 5% significance level	PRCCD, AJEXDI
ES3Y5	Expected Shortfall estimated over past 3 years at 5% significance level	PRCCD, AJEXDI
ES5Y5	Expected Shortfall estimated over past 5 years at 5% significance level	PRCCD, AJEXDI

Notes: This table reports the set of accounting- and market-based covariates that we use in our empirical analysis. The first column lists names of covariates, while the second column provides their respective definitions. Financial information and stock price information are sourced from the Compustat database, covering an analysis period from 1990 to 2016. The third column states the specific Compustat data items that we use to calculate respective covariates. Value-at-Risk (VAR) and Expected Shortfall (ES) are estimated at 1% and 5% significance levels using three-month, six-month, one-year, three-year and five-year daily returns.

Table 3: Descriptive Statistics

Variable	Descriptive Measures	Financially Distressed	Censored	Variable	Descriptive Measures	Financially Distressed	Censored
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
TLMTA				MB			
	Mean	0.3036	0.3436		Mean	3.1674	2.4524
	Median	0.2206	0.2965		Median	2.6238	2.0030
	SD	0.2462	0.2404		SD	2.0926	1.5761
	Minimum	0.0351	0.0351		Minimum	0.6588	0.6588
	Maximum	0.8445	0.8445		Maximum	6.6199	6.6199
CMTA				PRICE			
	Mean	0.1981	0.1130		Mean	0.9959	2.1294
	Median	0.1638	0.0649		Median	0.8458	2.6300
	SD	0.1505	0.1249		SD	0.8323	0.7968
	Minimum	0.0027	0.0027		Minimum	0.1177	0.1177
	Maximum	0.4510	0.4510		Maximum	2.7080	2.7080
NIMTA AVG				RSIZE			
	Mean	-0.1786	-0.0193		Mean	-11.9503	-10.2865
	Median	-0.1768	0.0191		Median	-12.1180	-10.3204
	SD	0.1042	0.0988		SD	1.2713	1.8230
	Minimum	-0.3038	-0.3038		Minimum	-13.4013	-13.4013
	Maximum	0.0750	0.0750		Maximum	-6.9217	-6.8132
EXRETA VG				FES			
	Mean	-0.0071	-0.0078		Mean	0.0459	0.0241
	Median	-0.0049	-0.0050		Median	0.0209	0.0111
	SD	0.0572	0.0429		SD	0.0509	0.0332
	Minimum	-0.0983	-0.0983		Minimum	0.0000	0.0000
	Maximum	0.0711	0.0711		Maximum	0.1289	0.1289
SIGMA				TMTA			
	Mean	0.8237	0.5526		Mean	0.0003	0.0115
	Median	0.7681	0.4733		Median	0.0000	0.0090
	SD	0.3079	0.2989		SD	0.0075	0.0158
	Minimum	0.1962	0.1962		Minimum	-0.0174	-0.0174
	Maximum	1.2851	1.2851		Maximum	0.0494	0.0494
VAR3M1				VAR3M5			
	Mean	0.1367	0.0878		Mean	0.0831	0.0543
	Median	0.1012	0.0671		Median	0.0663	0.0437
	SD	0.1178	0.0777		SD	0.0607	0.0422
	Minimum	1.0000	1.0000		Minimum	0.7153	1.0000
	Maximum	0.0033	0.0001		Maximum	0.0019	4.5e-05
VAR6M1				VAR6M5			
	Mean	0.1621	0.1057		Mean	0.0831	0.0556
	Median	0.1178	0.0797		Median	0.0685	0.0459
	SD	0.1384	0.0941		SD	0.0557	0.0398
	Minimum	1.0000	1.0000		Minimum	0.5836	1.0000
	Maximum	0.0036	8.2e-05		Maximum	0.0019	2.4e-06
VAR1Y1				VAR1Y5			
	Mean	0.1960	0.1241		Mean	0.0808	0.0545
	Median	0.1423	0.0921		Median	0.0687	0.0463

	SD	0.1662	0.1135		SD	0.0512	0.0361
	Minimum	1.0000	1.0000		Minimum	0.5886	1.0000
	Maximum	0.0109	4.2e-05		Maximum	0.0009	1.1e-05
VAR3Y1				VAR3Y5			
	Mean	0.2952	0.1687		Mean	0.0808	0.0537
	Median	0.2050	0.1202		Median	0.0717	0.0475
	SD	0.2367	0.1595		SD	0.0450	0.0321
	Minimum	1.0000	1.0000		Minimum	0.3717	1.0000
	Maximum	0.0481	0.0023		Maximum	0.0031	2.3e-05
VAR5Y1				VAR5Y5			
	Mean	0.3518	0.1945		Mean	0.0781	0.0529
	Median	0.2381	-0.1367		Median	0.0716	0.0477
	SD	0.2659	0.1830		SD	0.0393	0.0299
	Minimum	1.0000	1.0000		Minimum	0.3197	1.0000
	Maximum	0.0510	0.0011		Maximum	5.4e-05	2.6e-07
ES3M1				ES3M5			
	Mean	0.2142	0.1287		Mean	0.1326	0.0816
	Median	0.1348	0.0840		Median	0.0930	0.0608
	SD	0.2245	0.1555		SD	0.1186	0.0761
	Minimum	1.0000	1.0000		Minimum	1.0000	1.0000
	Maximum	0.0247	0.0007		Maximum	0.0047	7.2e-05
ES6M1				ES6M5			
	Mean	0.2323	0.1388		Mean	0.1325	0.0869
	Median	0.1561	0.0986		Median	0.0954	0.0648
	SD	0.2197	0.1404		SD	0.1163	0.0827
	Minimum	1	1		Minimum	1.0000	1.0000
	Maximum	0.0009	0.0001		Maximum	0.0084	2.4e-06
ES1Y1				ES1Y5			
	Mean	0.2833	0.1658		Mean	0.1239	0.0861
	Median	0.1924	0.1190		Median	0.0930	0.0656
	SD	0.2404	0.1568		SD	0.1134	0.0816
	Minimum	1.0000	1.0000		Minimum	1.0000	1.0000
	Maximum	0.0002	4.2e-05		Maximum	0.0034	5.4e-05
ES3Y1				ES3Y5			
	Mean	0.3775	0.2043		Mean	0.1151	0.0860
	Median	0.2876	0.1496		Median	0.0896	0.0672
	SD	0.2554	0.1775		SD	0.1090	0.0878
	Minimum	1.0000	1.0000		Minimum	1.0000	1.0000
	Maximum	0.0033	0.0000		Maximum	0.0002	2.3e-05
ES5Y1				ES5Y5			
	Mean	0.4033	0.2200		Mean	0.1150	0.0847
	Median	0.3045	0.1603		Median	0.0877	0.0666
	SD	0.2652	0.1896		SD	0.1185	0.0930
	Minimum	1.0000	1.0000		Minimum	1.0000	1.0000
	Maximum	0.0028	2.9e-05		Maximum	5.4e-05	2.6e-07

Notes: This table provides descriptive statistics for covariates listed in columns (1) and (5). To facilitate comparison, descriptive measures are reported separately for financially distressed (columns 3 and 7) and censored (columns 4 and 8) groups. SD is standard deviation.

Table 4: Correlation Matrix

Variable	1	2	3	4	5	6	7	8	9	10
Panel A: Correlation among Main Variables										
TLMTA (1)	1.0000									
CMTA (2)	-0.2325	1.0000								
ANIMTA (3)	-0.0290	-0.3093	1.0000							
AMEXRET (4)	-0.2127	-0.1358	0.2006	1.0000						
CSIGMA (5)	0.0742	0.1957	-0.4687	-0.2559	1.0000					
MB (6)	-0.3091	-0.1874	-0.1379	0.1980	0.0198	1.0000				
PRICE (7)	-0.2208	-0.2597	0.5218	0.3022	-0.6088	0.1081	1.0000			
RSIZE (8)	-0.1525	-0.2596	0.3514	0.2116	-0.5172	0.2353	0.7182	1.0000		
FES (9)	0.3232	-0.0709	-0.2893	-0.0848	0.1460	0.1666	-0.1716	-0.0575	1.0000	
TMTA (10)	-0.0890	-0.1177	0.4479	0.0752	-0.2527	-0.1184	0.2865	0.1743	-0.2397	1.0000
Panel B: Correlation among Main and Tail Risk Variables										
VAR3M1	-0.0860	-0.1368	0.3489	0.2263	-0.7607	-0.0195	0.4636	0.3784	-0.1142	0.1768
VAR6M1	-0.0743	-0.1441	0.3521	0.2774	-0.6806	-0.0341	0.4320	0.3654	-0.1042	0.1799
VAR1Y1	-0.0490	-0.1538	0.3482	0.2566	-0.5946	-0.0485	0.4035	0.3484	-0.1042	0.1797
VAR3Y1	-0.0166	-0.1604	0.3474	0.1556	-0.4454	-0.0835	0.3493	0.3092	-0.1253	0.2015
VAR5Y1	-0.0096	-0.1620	0.3427	0.1250	-0.3836	-0.0943	0.3341	0.2976	-0.1300	0.1997
VAR3M5	-0.1004	-0.1573	0.4021	0.2570	-0.8464	-0.0058	0.5315	0.4274	-0.1309	0.2031
VAR6M5	-0.1012	-0.1661	0.4210	0.3059	-0.8335	-0.0072	0.5475	0.4454	-0.1286	0.2134
VAR1Y5	-0.0858	-0.1746	0.4390	0.2813	-0.8154	-0.0179	0.5688	0.4677	-0.1306	0.2260
VAR3Y5	-0.0419	-0.1740	0.4525	0.1842	-0.7576	-0.0453	0.5664	0.4866	-0.1378	0.2545
VAR5Y5	-0.0049	-0.1803	0.4413	0.1511	-0.7144	-0.0579	0.5598	0.4961	-0.1275	0.2538
ES3M1	-0.0500	-0.1115	0.2676	0.1478	-0.5639	-0.0311	0.3263	0.2806	-0.0924	0.1423
ES6M1	-0.0667	-0.1336	0.3214	0.2064	-0.5859	-0.0330	0.3920	0.3392	-0.0994	0.1646
ES1Y1	-0.0543	-0.1514	0.3463	0.1939	-0.5485	-0.0371	0.4193	0.3713	-0.1116	0.1820
ES3Y1	-0.0185	-0.1747	0.4060	0.1491	-0.5130	-0.0851	0.4375	0.3889	-0.1460	0.2257
ES5Y1	-0.0042	-0.1760	0.3908	0.1331	-0.4524	-0.0981	0.4085	0.3632	-0.1499	0.2231
ES3M5	-0.0749	-0.1414	0.3582	0.2088	-0.7499	-0.0251	0.4491	0.3687	-0.1142	0.1777
ES6M5	-0.0718	-0.1337	0.3369	0.2546	-0.6519	-0.0254	0.4096	0.3444	-0.0935	0.1678
ES1Y5	-0.0503	-0.1346	0.3129	0.2390	-0.5811	-0.0280	0.3782	0.3143	-0.0823	0.1580
ES3Y5	-0.0083	-0.1067	0.2509	0.1199	-0.4184	-0.0413	0.2733	0.2345	-0.0713	0.1434
ES5Y5	0.0072	-0.0931	0.2046	0.0757	-0.3091	-0.0432	0.2137	0.1907	-0.0582	0.1177

Notes: This table reports correlation among the set of covariates. Panel A displays correlations among main variables suggested by Campbell *et al.* (2008) to which we supplemented FES and TMTA. Panel B provides correlations among main variables and the respective tail risk measures (VaR and ES).

Table 5: Univariate Regression

Variables	Coefficient	Standard Error	z-statistics	AME %	
(1)	(2)	(3)	(4)	(5)	(6)
TLMTA	1.5100 ^a	0.1471	10.26	0.47 ^a	8
CMTA	4.9993 ^a	0.2151	23.24	2.65 ^a	5
NIMTAAVG	-9.8945 ^a	0.2586	-38.25	-7.01 ^a	3
EXRETAVG	-19.7331 ^a	0.7401	-26.66	-12.46 ^a	2
SIGMA	2.0090 ^a	0.0946	21.22	1.33 ^a	----
MB	-0.1145 ^a	0.0218	-5.24	-0.04 ^a	9
PRICE	-1.9034 ^a	0.0515	-36.96	-1.44 ^a	6
RSIZE	-1.1209 ^a	0.0400	-27.96	-0.62 ^a	7
FES	11.1207 ^a	0.8059	13.80	5.08 ^a	4
TMTA	-53.9981 ^a	2.8121	-19.20	-30.06 ^a	1
AGE	0.4369 ^a	0.0437	9.99	0.13 ^a	----
INDRISK	18.5003 ^a	1.8566	9.96	6.93 ^a	----
VAR3M1	2.7077 ^a	0.2558	10.58	1.37 ^a	----
VAR6M1	2.1995 ^a	0.2175	10.11	1.13 ^a	----
VAR1Y1	1.9750 ^a	0.1822	10.84	1.03 ^a	----
VAR3Y1	2.0923 ^a	0.1366	15.31	1.19 ^a	----
VAR5Y1	2.2648 ^a	0.1268	17.86	1.33 ^a	----
VAR3M5	5.4376 ^a	0.4719	11.52	2.82 ^a	----
VAR6M5	6.0463 ^a	0.5137	11.77	3.18 ^a	----
VAR1Y5	7.3480 ^a	0.5778	12.72	3.89 ^a	----
VAR3Y5	11.3719 ^a	0.7018	16.20	5.93 ^a	----
VAR5Y5	13.8072 ^a	0.8320	16.59	7.15 ^a	----
ES3M1	1.3507 ^a	0.1318	10.24	0.67 ^a	----
ES6M1	1.7907 ^a	0.1418	12.62	0.94 ^a	----
ES1Y1	1.9188 ^a	0.1320	14.53	1.05 ^a	----
ES3Y1	2.4711 ^a	0.1221	20.23	1.55 ^a	----
ES5Y1	2.5243 ^a	0.1226	20.57	1.59 ^a	----
ES3M5	2.8589 ^a	0.2468	11.58	1.48 ^a	----
ES6M5	2.2348 ^a	0.2375	9.41	1.12 ^a	----
ES1Y5	1.8420 ^a	0.2478	7.43	0.90 ^a	----
ES3Y5	1.2138 ^a	0.2513	4.83	0.56 ^a	----
ES5Y5	1.2415 ^a	0.2363	-5.25	0.57 ^a	----

Notes: a [b] [c] significant at the 1% (5%) [10%] level (two-sided test). This table reports univariate regression estimates of respective covariates using our proposed definition of financial distress as the dependent variable. AME is average marginal effects in percentages. Column (6) reports the ranking of the main variables based on the magnitude of their average marginal effects (AME).

Table 6: Baseline Multivariate Model

Variables	B	SE	Z-statistics
(1)	(2)	(3)	(4)
TLMTA	0.4469 ^b	0.1885	2.3700
CMTA	3.5672 ^a	0.2759	12.930
NIMTAAVG	-6.8544 ^a	0.3733	-18.3600
EXRETAVG	-5.5647 ^a	0.8017	-6.9400
SIGMA	----	----	----
MB	----	----	----
PRICE	-1.1429 ^a	0.0553	-20.6500
RSIZE	----	----	----
FES	6.0002 ^a	0.8741	6.8600
TMTA	-18.0571 ^a	3.0427	-5.9300
AGE	0.2117 ^a	0.0655	3.2300
INDRISK	11.3057 ^a	2.4692	4.5800
Model's goodness of fit and prediction performance measures			
chi2	1835.3700 ^a	AUROC-W	0,9275
likelihood	-3688.0054	AUROC-H	0,9370
R ²	0.3917	N = 0+1	59,602
		N = 1	1,160

Notes: a (b) [c] significant at the 1% (5%) [10%] level (two-sided test). This table reports multivariate regression estimates that employ only the main variables discussed in Section 4.2. The baseline multivariate model is developed following the method suggested by Gupta *et al.* (2017), (see section 5). MB, RSIZE and SIGMA are excluded from the model for the reasons mentioned in section 5 (related to the expected coefficient sign and the p-value, or the strong correlation with other covariates). N = 0+1 represents the total number of observations for the two groups of firms between 1990 and 2016. N = 1 is the number of observations for the financially distressed firms. This table also reports the chi-square, the likelihood ratio and the coefficient of determination (McKelvey & Zavoina's R²) to measure the model's goodness of fit. AUROC-W is within-sample area under ROC curve and AUROC-H is for hold-out sample.

Figure 1: Area under ROC Curves

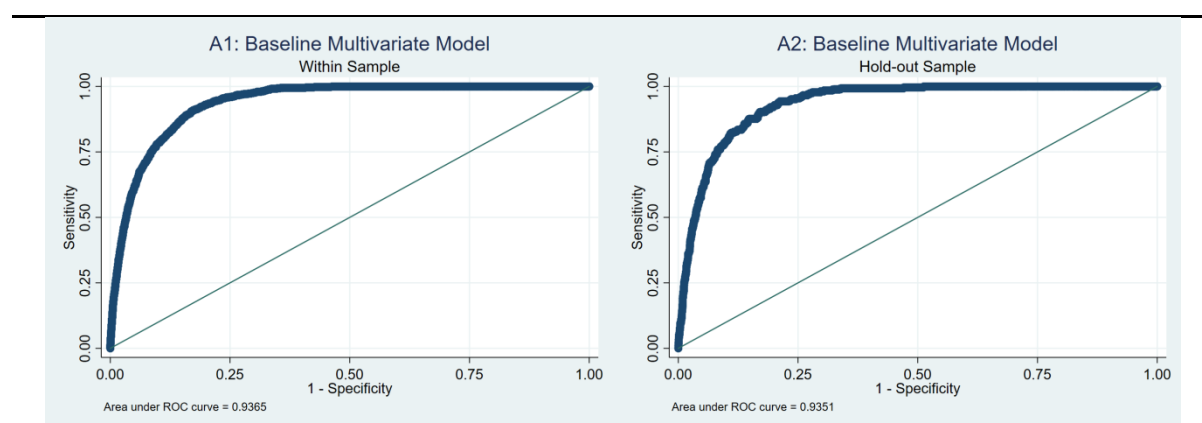


Table 7: Multivariate Regression Models with VaR

Variables	SIGMA	VAR3M1	VAR6M1	VAR1Y1	VAR3Y1	VAR5Y1	VAR3M5	VAR6M5	VAR1Y5	VAR3Y5	VAR5Y5
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Risk Measure											
β	0.3933 ^a	-0.3850	-0.2416	-0.1305	0.1827	0.4988 ^a	-0.4981	-0.7217	-0.5086	2.8675 ^a	5.4729 ^a
SE	0.1293	0.3679	0.3084	0.2573	0.1832	0.1621	0.6967	0.7649	0.8779	1.0326	1.2002
AME	0.4324 ^a	-0.4113	-0.2649	-0.1422	0.2007	0.5460 ^a	-0.5335	-0.7800	-0.5476	3.1140 ^a	5.8578 ^a
LR χ^2	9.1500 ^a	1.1200	0.6200	0.2600	0.9800	9.2200 ^a	0.5200	0.9000	0.3400	7.4400 ^a	20.0900 ^a
TLMTA											
β	1.6777 ^a	1.7287 ^a	1.6758 ^a	1.6716 ^a	1.7060 ^a	1.7024 ^a	1.7259 ^a	1.7193 ^a	1.6726 ^a	1.5198 ^a	1.4985 ^a
SE	0.1869	0.1900	0.1880	0.1881	0.1868	0.1866	0.1894	0.1898	0.1911	0.1928	0.1946
CMTA											
β	4.4137 ^a	4.2878 ^a	4.2929 ^a	4.2760 ^a	4.3262 ^a	4.3366 ^a	4.2772 ^a	4.2840 ^a	4.2926 ^a	4.4817 ^a	4.4909 ^a
SE	0.2765	0.2815	0.2790	0.2789	0.2763	0.2767	0.2811	0.2822	0.2835	0.2835	0.2862
NIMTA AVG											
β	-8.8183 ^a	-9.4454 ^a	-9.3537 ^a	-9.3560 ^a	-9.2265 ^a	-9.0873 ^a	-9.3721 ^a	-9.4359 ^a	-9.3950 ^a	-8.9950 ^a	-8.8013 ^a
SE	0.3887	0.3766	0.3738	0.3728	0.3655	0.3639	0.3794	0.3835	0.3862	0.3801	0.3796
EXRETA VG											
β	-8.4205 ^a	-8.9061 ^a	-8.9833 ^a	-8.8568 ^a	-8.6157 ^a	-8.5480 ^a	-8.9385 ^a	-8.8770 ^a	-8.9865 ^a	-8.2952 ^a	-8.4560 ^a
SE	0.8116	0.8258	0.8210	0.8193	0.8139	0.8100	0.8229	0.8265	0.8311	0.8342	0.8346
MB											
β	----	----	----	----	----	----	----	----	----	----	----
SE	----	----	----	----	----	----	----	----	----	----	----
PRICE											
β	----	----	----	----	----	----	----	----	----	----	----
SE	----	----	----	----	----	----	----	----	----	----	----
RSIZE											
β	----	----	----	----	----	----	----	----	----	----	----
SE	----	----	----	----	----	----	----	----	----	----	----
FES											
β	5.6236 ^a	5.4946 ^a	5.6969 ^a	5.6422 ^a	5.5770 ^a	5.4887 ^a	5.4792 ^a	5.5304 ^a	5.6726 ^a	5.8213 ^a	5.6669 ^a
SE	0.8511	0.8666	0.8567	0.8570	0.8540	0.8566	0.8650	0.8648	0.8700	0.8801	0.8942
TMTA											
β	-22.1676 ^a	-22.2201 ^a	-22.3155 ^a	-22.4146 ^a	-22.5008 ^a	-22.3727 ^a	-22.3495 ^a	-22.1072 ^a	-22.2728 ^a	-21.9623 ^a	-22.1098 ^a
SE	2.9598	2.9792	2.9707	2.9736	2.9595	2.9681	2.9759	2.9846	2.9979	3.0049	3.0251 ^a
AGE											
β	0.1274 ^a	0.0849 ^a	0.09023 ^a	0.09051 ^a	0.1015 ^a	0.1105 ^a	0.0850 ^a	0.0837 ^a	0.0811 ^a	0.1250 ^a	0.1705 ^a
SE	0.0620	0.0621	0.06153	0.06155	0.0612	0.0614	0.0621	0.0622	0.0629	0.0640	0.0658
RISK											
β	17.1060 ^a	16.1372 ^a	16.2123 ^a	16.0175 ^a	16.1081 ^a	15.8947 ^a	16.3399 ^a	16.0984 ^a	16.6254 ^a	17.2413 ^a	17.7536 ^a

SE	2.3778	2.3905	2.3698	2.3736	2.3526	2.3555	2.3876	2.3961	2.4021	2.4076	2.4023
Model's goodness of fit and prediction performance measures											
chi2	1995.7500 ^a	1940.4500 ^a	-3850.2880 ^a	-3863.4252 ^a	-3913.1469 ^a	-3910.7813 ^a	-3842.8243 ^a	-3809.9553 ^a	-3765.4383 ^a	-3710.9791 ^a	-3661.9754 ^a
likelihood	-3916.6505	-3820.1015	1979.6500	1971.2700	1992.3500	1986.4100	1945.7400	1947.4600	1935.7100	1915.1000	1868.5800
R ²	0.3330	0.3278	0.3309	0.3291	0.3303	0.3312	0.3279	0.3291	0.3295	0.3279	0.3293
N = 0+1	59,514	58,574	58,807	58,976	59,301	59,331	58,784	58,507	58,101	57,387	57,010
N = 1	1,155	1,123	1,133	1,135	1,153	1,155	1,129	1,118	1,107	1,087	1,070

Notes: a (b) [c] significant at the 1% (5%) [10%] level (two-sided test). This table reports multivariate regression estimates obtained by supplementing the baseline model reported in Table 6 with SIGMA and VaR measures. Estimates of the risk measures as listed in columns (2) to (12) are provided in rows (4) to (7). Column (2) reports regression estimates with SIGMA being the risk measure. Columns (3) to (7) report respective multivariate regression estimates supplementing VaR as the risk measure (three-month, six-month, one-year, three-year and five-year at the 99% confidence interval) and columns (7) to (12) report the same at 1% confidence interval. The chi-square for likelihood ratio test (LR χ^2) reported in row (7) of column (1) tests the existence of a significant difference between the log likelihood of the baseline model supplemented with risk variable, and log likelihood of baseline model. The last 5 rows of this table provide goodness of fit measures. Chi-square (χ^2), likelihood ratio, and the coefficient of determination (McKelvey & Zavoina's R^2) of respective multivariate models with respective risk variables are reported in the first three of the last five rows. N = 0+1 represents the total number of observations for the two groups of firms between 1990 and 2016. N = 1 is the number of observations of the financially distressed firms. PRICE is excluded from the model because of its moderate correlation with risk measures.

Table 8: Multivariate Regression Models with ES

Variables	SIGMA	ES3M1	ES6M1	ES1Y1	ES3Y1	ES5Y1	ES3M5	ES6M5	ES1Y5	ES3Y5	ES5Y5
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Risk Measure											
β	0.3933 ^a	0.0795	0.1453	0.3743 ^b	0.5921 ^a	0.7179 ^a	-0.1104	-0.3087	-0.4542	-0.3866	0.1676
SE	0.1293	0.1734	0.1914	0.1717	0.1647	0.1598	0.3388	0.3394	0.3559	0.3617	0.3040
AME	0.4324 ^a	0.0867	0.1601	0.4085 ^b	0.6506 ^a	0.7845 ^a	-0.1204	-0.3348	-0.4941	-0.4168	0.1787
LR χ^2	9.1500 ^a	0.2100	0.5700	4.6300 ^b	12.5000 ^a	19.5300 ^a	0.1100	0.8400	1.6900	1.1900	0.3000
TLMTA											
β	1.6777 ^a	1.7262 ^a	1.6636 ^a	1.6758 ^a	1.6810 ^a	1.6970 ^a	1.7278 ^a	1.7263 ^a	1.7430 ^a	1.6365 ^a	1.7042 ^a
SE	0.1869	0.1873	0.1869	0.1874	0.1861	0.1862	0.1874	0.1879	0.1889	0.1904	0.1917
CMTA											
β	4.4137 ^a	4.3306 ^a	4.3297 ^a	4.3348 ^a	4.3297 ^a	4.3186 ^a	4.3266 ^a	4.3006 ^a	4.3886 ^a	4.3474 ^a	4.3669 ^a
SE	0.2765	0.2768	0.2765	0.2776	0.2757	0.2763	0.2769	0.2777	0.2785	0.2808	0.2829
NIMTA AVG											
β	-8.8183 ^a	-9.2858 ^a	-9.1883 ^a	-9.0785 ^a	-8.9590 ^a	-8.9260 ^a	-9.3458 ^a	-9.3954 ^a	-9.4179 ^a	-9.3566 ^a	-9.2940 ^a
SE	0.3887	0.3632	0.3674	0.3685	0.3679	0.3662	0.3682	0.3696	0.3680	0.3656	0.3662
EXRETA VG											
β	-8.4205 ^a	-8.6351 ^a	-8.8500 ^a	-8.6627 ^a	-8.4470 ^a	-8.4491 ^a	-8.6741 ^a	-8.7301 ^a	-8.6136 ^a	-8.5953 ^a	-8.6407 ^a
SE	0.8116	0.8127	0.8144	0.8157	0.8101	0.8080	0.8126	0.8152	0.8165	0.8295	0.8304
MB											
β	----	----	----	----	----	----	----	----	----	----	----
SE	----	----	----	----	----	----	----	----	----	----	----
PRICE											
β	----	----	----	----	----	----	----	----	----	----	----
SE	----	----	----	----	----	----	----	----	----	----	----
RSIZE											
β	----	----	----	----	----	----	----	----	----	----	----
SE	----	----	----	----	----	----	----	----	----	----	----
FES											
β	5.6236 ^a	5.5976 ^a	5.5690 ^a	5.5714 ^a	5.4286 ^a	5.3207 ^a	5.6089 ^a	5.6439 ^a	5.6865 ^a	5.8633 ^a	5.5143 ^a
SE	0.8511	0.8534	0.8523	0.8553	0.8531	0.8566	0.8531	0.8555	0.8593	0.8699	0.8782
TMTA											
β	-22.1676 ^a	-22.5036 ^a	-22.4534 ^a	-22.4892 ^a	-22.3682 ^a	-22.2133 ^a	-22.5895 ^a	-22.3922 ^a	-22.3892 ^a	-22.0471 ^a	-22.5668 ^a
SE	2.9598	2.9603	2.9567	2.9683	2.9660	2.9749	2.9598	2.9632	2.9661	2.9939	3.0034
AGE											
β	0.1274 ^b	0.0958	0.0967	0.0992	0.1161 ^c	0.1207 ^b	0.0926	0.0947	0.0890	0.0818	0.0948
SE	0.0620	0.0612	0.0610	0.0613	0.0614	0.0615	0.0612	0.0613	0.0616	0.0623	0.0626
RISK											
β	17.1060 ^a	16.0585 ^a	2.3570 ^a	16.4370 ^a	16.1974 ^a	16.0815 ^a	15.9736 ^a	15.9074 ^a	15.7392 ^a	16.2860 ^a	16.8044 ^a

SE	2.3778	2.3568	0.2099	2.3635	2.3534	2.3535	2.3585	2.3634	2.3728	2.3852	2.3850
Model's goodness of fit and prediction performance measures											
chi2	1995.7500 ^a	1987.8300 ^a	1996.2000 ^a	-3889.0025 ^a	-3910.4240 ^a	-3909.6051 ^a	1988.3900 ^a	1978.6600 ^a	1972.8200 ^a	1938.1800 ^a	1900.8100 ^a
likelihood	-3916.6505	-3915.4802	-3903.2740	1978.4800	1992.8400	1983.9100	-3915.5515	-3892.8447	-3866.3157	-3784.0133	-3765.6682
R ²	0.3330	0.3294	0.3304	0.3298	0.3314	0.3320	0.3297	0.3290	0.3297	0.3259	0.3270
N = 0+1	59,514	59,362	59,254	59,253	59,437	59,458	59,388	59,268	58,858	58,201	57,965
N = 1	1,155	1,154	1,149	1,143	1,153	1,155	1,154	1,145	1,140	1,108	1,100

Notes: a (b) [c] significant at the 1% (5%) [10%] level (two-sided test). This table reports multivariate regression estimates obtained by supplementing the baseline model reported in Table 6 with SIGMA and ES measures. Estimates of the risk measures as listed in columns (2) to (12) are provided in rows (4) to (7). Column (2) reports regression estimates with SIGMA being the risk measure. Columns (3) to (7) report respective multivariate regression estimates supplementing ES as the risk measure (three-month, six-month, one-year, three-year and five-year at the 99% confidence interval) and columns (7) to (12) report the same at 1% confidence interval. The chi-square for likelihood ratio test (LR χ^2) reported in row (7) of column (1) tests the existence of a significant difference between the log likelihood of the baseline model supplemented with risk variable, and log likelihood of baseline model. The last 5 rows of this table provide goodness of fit measures. Chi-square (Chi2), likelihood ratio and the coefficient of determination (McKelvey & Zavoina's R²) of respective multivariate models with respective risk variables are reported in the first three of the last five rows. N = 0+1 represents the total number of observations for the two groups of firms between 1990 and 2016. N = 1 is the number of observations of the financially distressed firms. PRICE is excluded from the model because of its moderate correlation with risk measures.